# PAPER_710: NGC 2014 + NGC 2020: Tapestry of Blazing Starbirth UQFF

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## Abstract

NGC 2014 (red, HII emission) and NGC 2020 (blue, WR-dominated) form the iconic Tapestry of Blazing Starbirth in the Large Magellanic Cloud (163,000 ly). They represent a dual-arm star-forming complex driven by Wolf-Rayet and OB stellar winds with rapid secondary star formation. The UQFF MUGE is derived for this system.

## 1. System Parameters

| Parameter | Value | Description |
|---|---|---|
| $M_0$ | $240 M_\odot$ | Initial OB/WR stellar mass |
| $r_{region}$ | $9.461 \times 10^{16}$ m | Region span (10 ly) |
| $\dot{M}_{SF}$ | 41.67 | Secondary SF mass fraction |
| $\tau_{SF}$ | $1.578 \times 10^{14}$ s | SF timescale (5 Myr) |
| $B_{LMC}$ | $10^{-6}$ T | LMC magnetic field |

## 2. Master UQFF Gravity Equation

$$g_{TB}(t) = \frac{GM(t)}{r^2}(1+H_0 t)\left(1 - \frac{B}{B_{crit}}\right) + (U_{g1}+U_{g4})(1+f_{TRZ}) + \frac{\Lambda c^2}{3} + q(v \times B)\left(1 + \frac{\rho_{UA}}{\rho_{SCm}}\right) \times 10^{-}$$

### 2.1 Secondary Star Formation (WR + OB triggered)

$$M(t) = M_0(1 + 41.67\, e^{-t/\tau_{SF}})$$

At $t = 2.5$ Myr: $M \approx 240\,(1 + 41.67 \cdot 0.709) \approx 240 \times 30.7 \approx 7368 M_\odot$

## 3. Numerical Result

$$g_{TB}(t = 2.5 \text{ Myr}) \approx 1.053 \times 10^{-4}\,\text{m/s}^2$$

## References

- Hubble WFC3 image (2020)
- Crowther (2007), ARA&A, 45, 177 (Wolf-Rayet stellar physics)
- UQFF Framework, Star-Magic Session 175

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